

PROJECT DOCUMENTATION

Azure Batch for an Engineering Simulation Workload

Cloud-based parallel processing prototype

Project Information	Details
Institution	Koneru Lakshmaiah Education Foundation (KLEF)
Department	Department of Computer Science and Engineering
Team	T177
Project Theme	Azure for Students • Azure Batch • 300 simulation tasks

Team Member	Student ID
Ch. Sri Neha	2400031988
A. Keerthi	2400032168
B. Shavya Sri	2400032179
K. Sri Sai Pavan	2400032315

Document Scope

This documentation is based on the final project presentation and records the project introduction, problem statement, objectives, SimFlow prototype, Azure Batch architecture, Azure resources, compute pool, Batch job, task submission, execution monitoring, final results, limitations, conclusion, and future scope.

The project presentation keeps the standalone SimFlow prototype and the Azure Batch implementation separate. SimFlow demonstrates the engineering application concept, while Azure Batch demonstrates cloud batch processing.

The numerical workload is a demonstration workload rather than a production engineering solver. The demonstrated results therefore represent workload execution and completion rather than physical engineering predictions.

Table of Contents

1. Introduction
2. Problem Statement
3. Project Objectives
4. Engineering Use Case / Prototype
5. Prototype Simulation
6. Proposed Azure Architecture
7. Azure Resources Used
8. Azure Batch Account
9. Compute Pool Configuration
10. Batch Job
11. Task Submission
12. Execution Monitoring
13. Final Task Results
14. Prototype and Azure Implementation
15. Advantages and Applications
16. Limitations and Cost Considerations
17. Conclusion
18. Future Scope
19. Appendix: Screenshot Evidence

1. Introduction

Engineering simulations often require many independent runs with different input parameters. Running a large set of independent calculations on a single machine can be time-consuming.

Azure Batch provides a managed way to create compute pools, jobs and tasks, and schedule those tasks across compute nodes. This project demonstrates that workflow using a controlled numerical simulation workload.

A separate SimFlow Workload prototype demonstrates the application-side concept through inputs, scenarios, processing and completion.

2. Problem Statement

A simulation study may require hundreds of independent scenarios or parameter combinations. Sequential execution can increase turnaround time and makes scaling difficult.

- A large engineering workload may contain hundreds of independent simulation cases.
- Sequential processing can increase overall completion time.
- Independent tasks can be distributed across a small pool of compute nodes.
- Azure Batch can schedule queued tasks onto available compute nodes.

3. Project Objectives

- Create an Azure Batch account and a compute pool.
- Create a Batch job for the simulation workload.
- Generate and submit 300 independent simulation tasks.
- Monitor task execution and verify completion.
- Demonstrate how the same pattern can support larger engineering simulation studies.

4. Engineering Use Case / Prototype

Stage	Prototype Representation
Inputs	Number of simulations according to user requirements
Scenarios	300 independent simulation cases
Processing	Numerical simulation for each case
Results	Completion of jobs as requested

The prototype is a standalone conceptual workload called SimFlow Workload. It demonstrates the engineering workload concept and does not need to connect to Azure for the stated project scope.

Proposed Azure Batch Architecture

Azure Batch Architecture for Engineering Simulation Workload

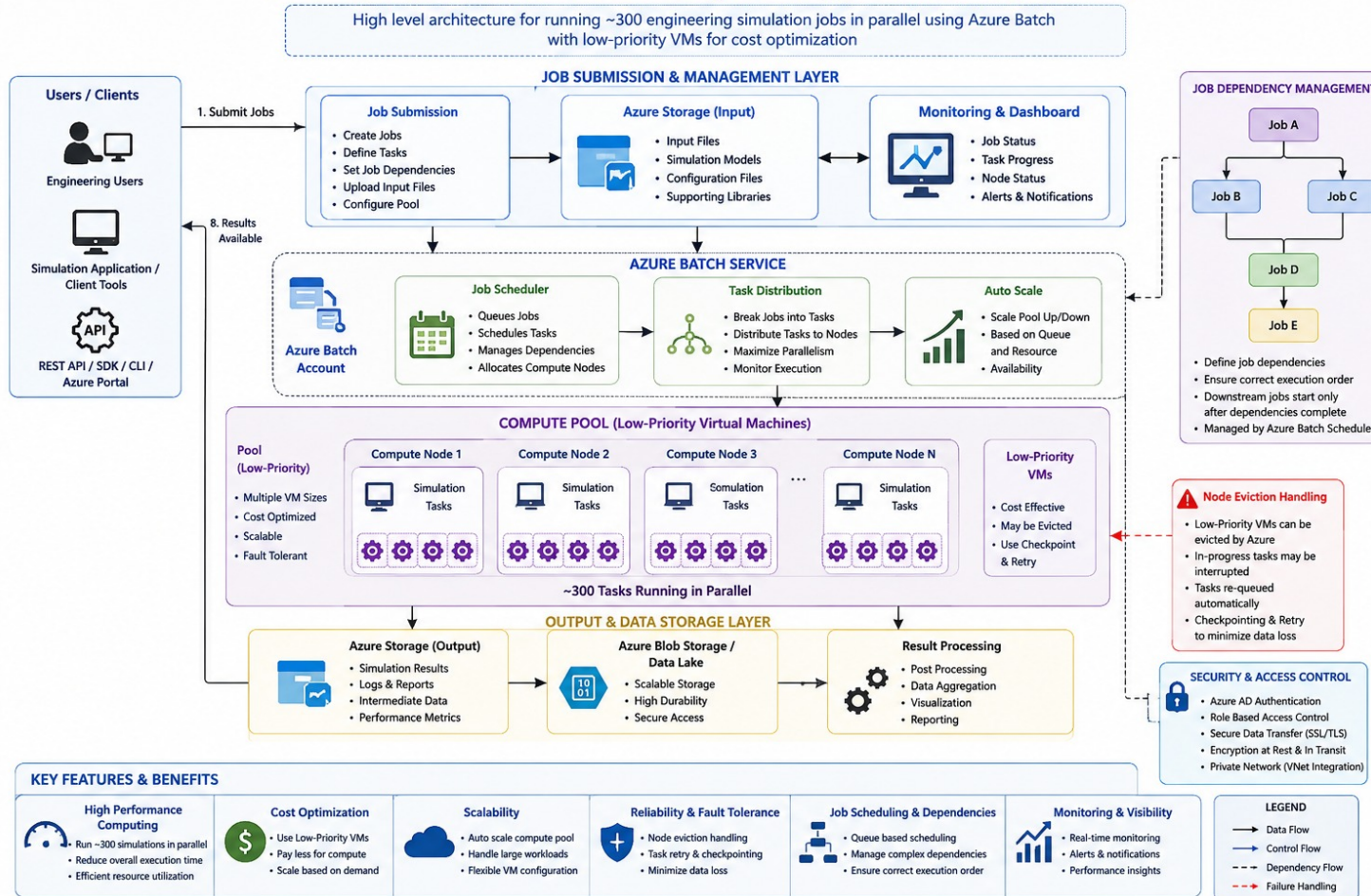


Figure 3. Azure Batch architecture for the engineering simulation workload.

Architecture Description

- 1. Job Submission and Management Layer:** Engineering users or client tools submit simulation jobs. The job submission component creates jobs, defines tasks, sets dependencies, uploads input files, and configures the compute pool. Azure Storage holds input files, simulation models, configuration files, and supporting libraries. Monitoring and Dashboard provides job status, task progress, node status, and alerts.
- 2. Azure Batch Service:** The Azure Batch account manages the workload through job scheduling, task distribution, and pool scaling. The Job Scheduler queues jobs and schedules tasks, while Task Distribution breaks the workload into individual tasks and assigns them to available compute nodes. Auto Scale can adjust the pool according to queue and resource requirements.
- 3. Compute Pool:** The compute pool provides the virtual machines where simulation tasks are executed. The architecture represents approximately 300 simulation tasks running in parallel across compute nodes. The design also shows low-priority virtual machines as a cost-optimization option, with checkpointing and retry used to reduce the effect of possible node eviction.
- 4. Job Dependency Management:** Dependencies define the correct execution order when one job or task must wait for another. For example, Job A can lead to Jobs B and C, followed by Job D and then Job E. This ensures downstream work starts only after the required preceding work is complete.

7. Project Modules

The project is organized into the following functional modules. Together they describe the application-side prototype and the Azure Batch execution workflow from workload definition through task verification.

Module	Purpose / Responsibility
1. SimFlow Prototype Module	Represents the engineering simulation workload at the application level. It demonstrates inputs, scenarios, processing, workload status, compute-node information, and results as a standalone conceptual prototype.
2. Azure Resource Management Module	Manages the Azure resources used by the implementation, including the Azure subscription, resource group, Batch account, and regional configuration.
3. Compute Pool Module	Provides the virtual machines used to execute simulation tasks. The documented implementation uses simulation-pool with two dedicated Standard_D2s_v3 nodes.
4. Batch Job Management Module	Creates and manages the logical Batch job. The project uses simulation-job, linked to simulation-pool, to group the simulation tasks.
5. Simulation Task Submission Module	Creates and submits the individual simulation workload. The project contains simulation-task-001 through simulation-task-300, submitted in three batches of 100.
6. Task Scheduling & Execution Module	Handles the queuing, scheduling, and execution of independent tasks on available compute nodes in the Batch pool.
7. Execution Monitoring Module	Monitors task progress and workload status through the Azure Portal and verifies the completion state of the submitted workload.
8. Results & Verification Module	Verifies final task states and execution evidence. The documented evidence shows tasks 284–300 as Completed with exit code 0.

Module Flow: SimFlow Prototype → Azure Resource Management → Azure Batch Account → Compute Pool → Batch Job → Simulation Tasks → Task Scheduling & Execution → Execution Monitoring → Results & Verification.

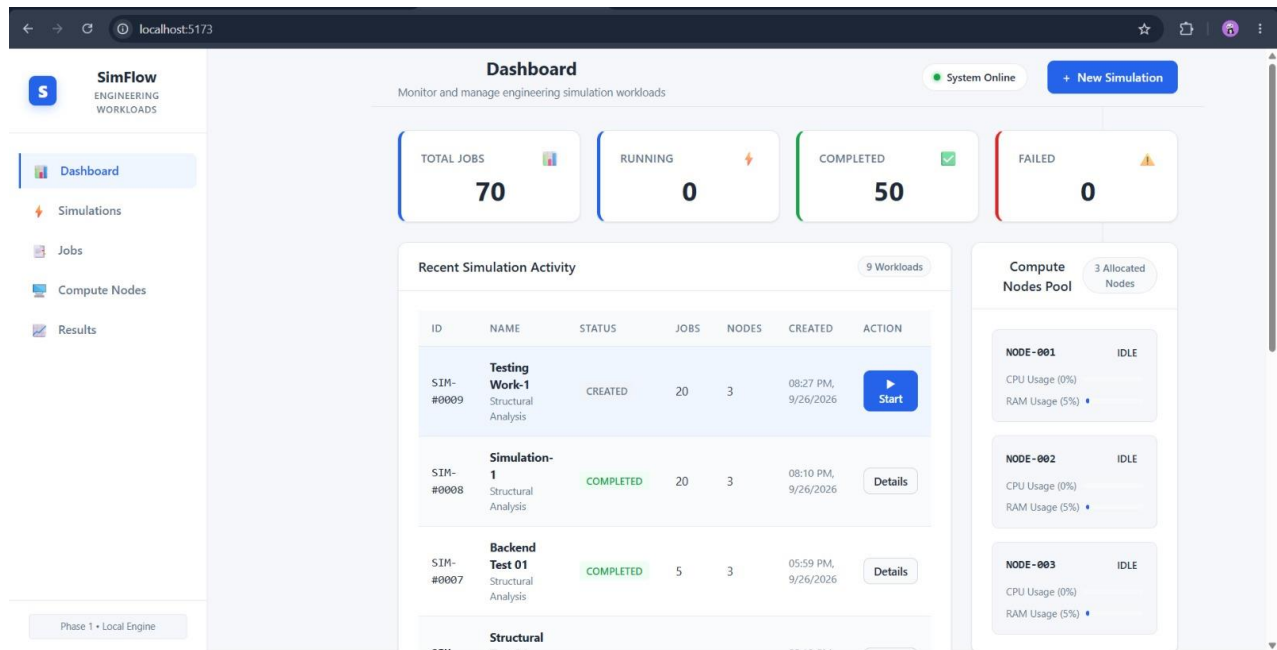


Figure 1. SimFlow prototype dashboard from the final presentation.

5. Prototype Simulation

The SimFlow prototype provides a visual representation of engineering simulation workloads, including workload status and compute-node information. It demonstrates the application concept independently of the Azure infrastructure.

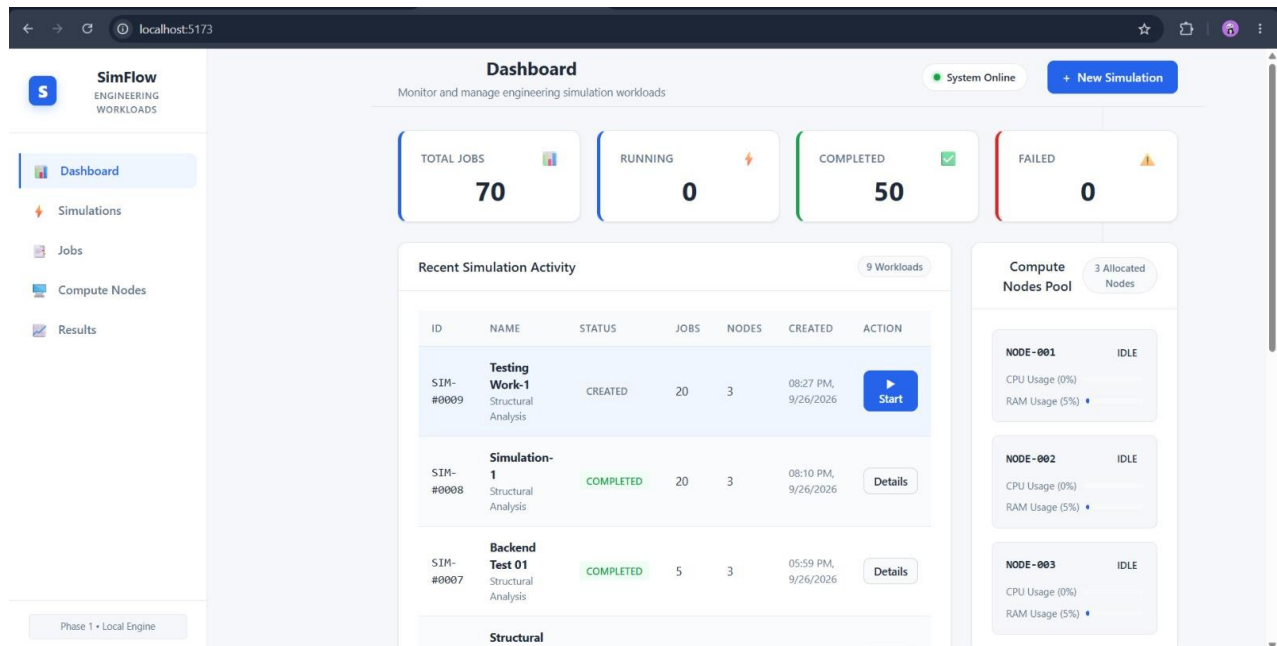


Figure 2. SimFlow workload monitoring view.

6. Proposed Azure Architecture

The proposed architecture uses Azure Portal or Cloud Shell to create and manage resources. An Azure Batch account contains a compute pool. A Batch job is associated with the pool, and the job contains 300 independent simulation tasks.

Component	Configuration
Azure Batch Account	engsimbatch2026neha
Compute Pool	simulation-pool
VM Size	Standard_D2s_v3
Dedicated Nodes	2
Batch Job	simulation-job
Tasks	simulation-task-001 through simulation-task-300

7. Azure Resources Used

Resource	Value
Subscription	Azure for Students
Resource Group	RG-Engineering-Simulation
Batch Account	engsimbatch2026neha
Region	India South Central
Pool	simulation-pool
VM Size	Standard_D2s_v3
Dedicated Nodes	2
Batch Job	simulation-job
Tasks	300

8. Azure Batch Account

The actual Azure implementation used the Batch account named engsimbatch2026neha. The Azure Portal evidence shows the account online in India South Central under the Azure for Students subscription and RG-Engineering-Simulation resource group.

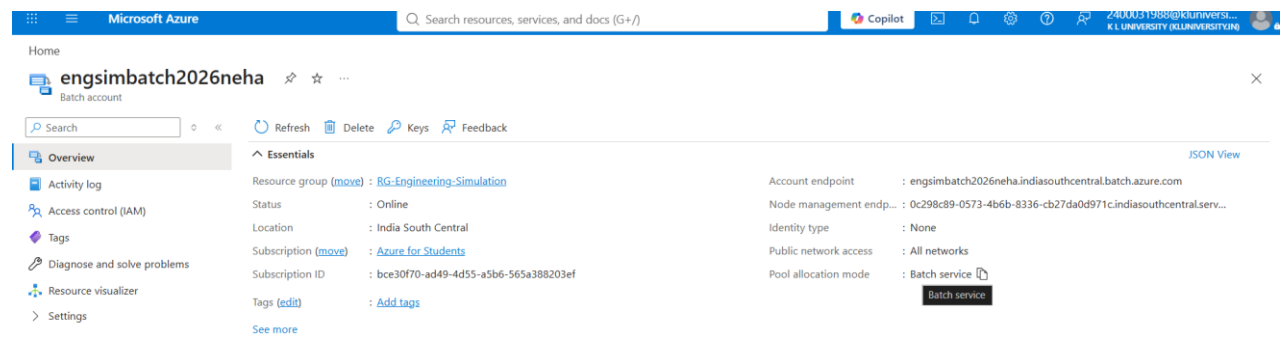


Figure 3. Azure Batch account overview.

9. Compute Pool Configuration

The actual workload used simulation-pool with two dedicated nodes. The pool used Standard_D2s_v3 virtual machines, providing four vCPUs in total. The pool was shown in a Steady allocation state.

Property	Value
Pool ID	simulation-pool
Dedicated nodes	2
Spot/low-priority nodes	0
Current vCPUs	4
VM size	STANDARD_D2S_V3
Allocation state	Steady

The screenshot shows the Azure Batch Pools configuration page for the account 'engsimbatch2026neha'. The left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Features, Applications, Pools (selected), Jobs, Job schedules, Monitoring, Automation, and Help. The main content area displays a table of pools. The table has columns: Pool ID, Dedicated nodes, Spot/low-priority nodes, Current vCPUs, VM size, and Allocation state. A single pool is listed: 'simulation-pool' with 2 dedicated nodes, 0 spot/low-priority nodes, 4 current vCPUs, VM size 'STANDARD_D2S_V3', and an allocation state of 'Steady' (indicated by a green checkmark). Above the table, there are filters (State == all), a search bar, and pagination controls (Pagination effort limit: 1, Actual: 1).

Figure 4. Azure Batch pool configuration.

10. Batch Job

A Batch job named simulation-job was created and linked to simulation-pool. The job provides the logical grouping for the 300 independent simulation tasks.

Property	Value
Job ID	simulation-job
State shown	Active
Linked pool	simulation-pool

The screenshot shows the Azure Batch Jobs view for the account 'engsimbatch2026neha'. The left sidebar is similar to the previous figure, with 'Jobs' selected. The main content area displays a table of jobs. The table has columns: ID, State, Pool, and Created. A single job is listed: 'simulation-job' with a state of 'Active' (indicated by a green checkmark), linked to the 'simulation-pool', and created on '26 Sept 2026, 22:56:55'. Above the table, there are filters (State == all), a search bar, and pagination controls (Pagination effort limit: 1, Actual: 1).

Figure 5. Azure Batch Jobs view.

11. Task Submission

The workload consists of 300 independent simulation tasks named simulation-task-001 through simulation-task-300. Each task represents a repeated numerical calculation performed independently.

- Tasks were submitted in three batches of 100 tasks.
- Each task used a non-admin task identity on the Ubuntu compute nodes.
- Two dedicated nodes were used to keep the educational demonstration small and cost-conscious.
- Azure Batch placed queued tasks onto available compute nodes.

Task	State	Created	Exit code
simulation-task-001	Completed	26 Sept 2026, 23:09:02	0
simulation-task-002	Completed	26 Sept 2026, 23:09:02	0
simulation-task-003	Completed	26 Sept 2026, 23:09:02	0
simulation-task-004	Completed	26 Sept 2026, 23:09:02	0
simulation-task-005	Completed	26 Sept 2026, 23:09:02	0
simulation-task-006	Completed	26 Sept 2026, 23:09:02	0
simulation-task-007	Completed	26 Sept 2026, 23:09:02	0
simulation-task-008	Completed	26 Sept 2026, 23:09:02	0
simulation-task-009	Completed	26 Sept 2026, 23:09:02	0
simulation-task-010	Completed	26 Sept 2026, 23:09:02	0
simulation-task-011	Completed	26 Sept 2026, 23:09:02	0

Figure 6. Azure Batch task monitoring and completion progress.

12. Execution Monitoring

Azure Portal was used to monitor the workload. The final task-list evidence shows an approximate task count of 300 and a completion indicator of 100% (300/300 completed).

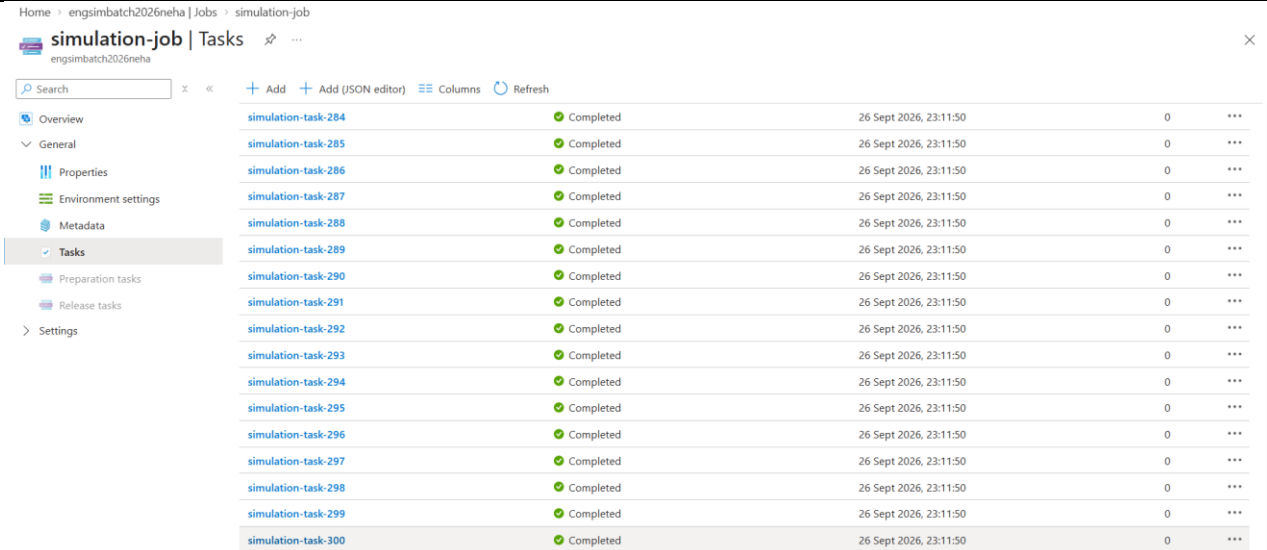
Task	State	Created	Exit code
simulation-task-001	Completed	26 Sept 2026, 23:09:02	0
simulation-task-002	Completed	26 Sept 2026, 23:09:02	0
simulation-task-003	Completed	26 Sept 2026, 23:09:02	0
simulation-task-004	Completed	26 Sept 2026, 23:09:02	0
simulation-task-005	Completed	26 Sept 2026, 23:09:02	0
simulation-task-006	Completed	26 Sept 2026, 23:09:02	0
simulation-task-007	Completed	26 Sept 2026, 23:09:02	0
simulation-task-008	Completed	26 Sept 2026, 23:09:02	0
simulation-task-009	Completed	26 Sept 2026, 23:09:02	0
simulation-task-010	Completed	26 Sept 2026, 23:09:02	0
simulation-task-011	Completed	26 Sept 2026, 23:09:02	0

Figure 7. Azure Batch execution monitoring.

13. Final Task Results

The final presentation includes task evidence for tasks 284–300. These tasks are displayed as Completed and show exit code 0.

Metric	Reported Value
Simulation tasks submitted	300
Dedicated compute nodes	2
Tasks completed	100% shown
Displayed final tasks	284–300
Displayed exit code	0



The screenshot shows the 'simulation-job | Tasks' view in the Azure Batch console. A left-hand navigation pane includes links for Overview, General, Properties, Environment settings, Metadata, Tasks (selected), Preparation tasks, Release tasks, and Settings. The main pane displays a table of tasks:

Task ID	Status	Time	Exit Code	Details
simulation-task-284	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-285	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-286	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-287	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-288	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-289	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-290	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-291	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-292	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-293	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-294	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-295	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-296	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-297	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-298	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-299	Completed	26 Sept 2026, 23:11:50	0	...
simulation-task-300	Completed	26 Sept 2026, 23:11:50	0	...

Figure 8. Final task evidence for tasks 284–300.

14. Prototype and Azure Implementation

Prototype: SimFlow Workload	Azure Implementation
Demonstrates engineering application concept.	Demonstrates cloud batch processing.
Shows inputs, scenarios, processing and results.	Azure Batch account + compute pool.
Standalone conceptual prototype.	Batch job + 300 independent tasks.
Does not need to connect to Azure for the defined scope.	Monitoring and completion verification.

Keeping these components separate allows the project to demonstrate both the application concept and the cloud batch-processing workflow without presenting the prototype as an Azure-connected engineering solver.

15. Advantages and Applications

- Parallel task execution: independent simulation cases can be scheduled across compute nodes.
- Scalability: the Batch pattern can be expanded by increasing compute resources when appropriate.
- Managed scheduling: Azure Batch handles task queuing and assignment to available nodes.

- Potential applications include structural analysis, thermal simulations, manufacturing parameter sweeps, aerospace studies, and other independent numerical workloads.

16. Limitations and Cost Considerations

16.1 Limitations

- The numerical workload is a demonstration workload, not a production engineering solver.
- The prototype and Azure implementation are separate.
- Demonstration results should not be interpreted as physical engineering predictions.

16.2 Cost / Resource Approach

- Azure for Students subscription was used.
- A small two-node pool was selected instead of hundreds of nodes.
- The configuration was sufficient to demonstrate Batch scheduling while limiting resource usage.

17. Conclusion

An Azure Batch environment was created for an engineering simulation workload. A compute pool with two Standard_D2s_v3 nodes was configured, a Batch job was created, and 300 independent tasks were submitted.

Azure Portal evidence shows 300 tasks at 100% completion, with the displayed final tasks returning exit code 0. The project demonstrates the basic cloud workflow for scaling independent simulation computations.

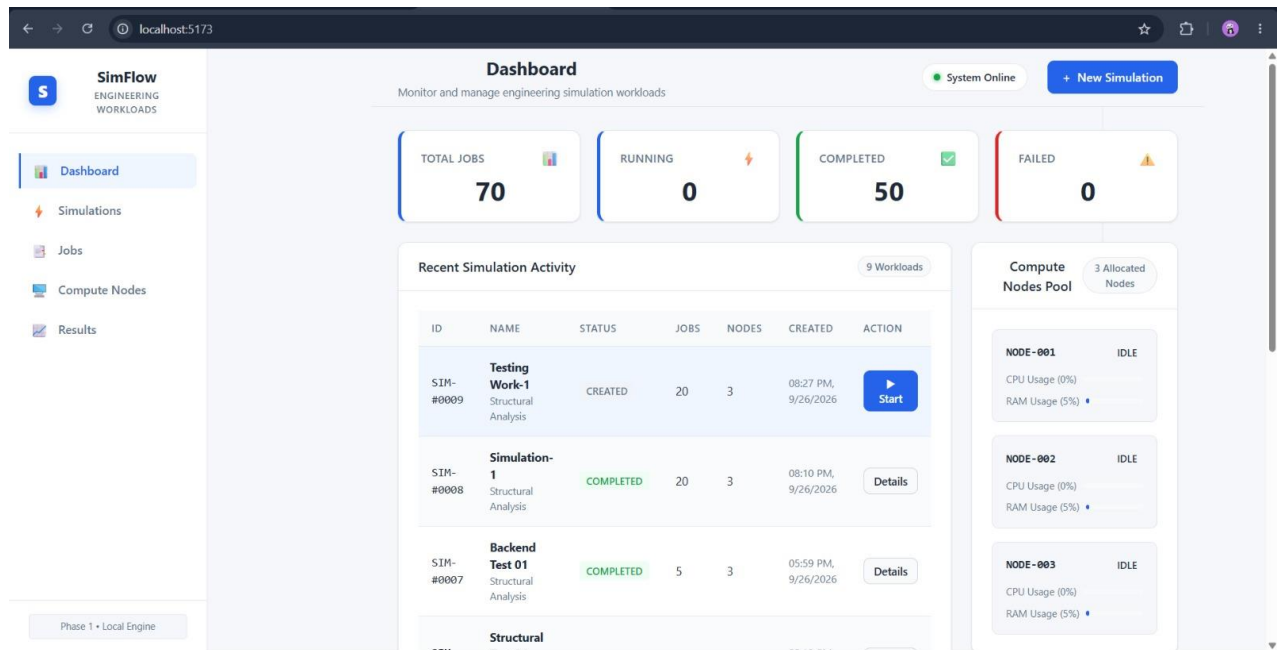
The SimFlow prototype complements the Azure work by demonstrating the standalone engineering workload concept, while Azure Batch demonstrates the distributed cloud execution workflow.

18. Future Scope

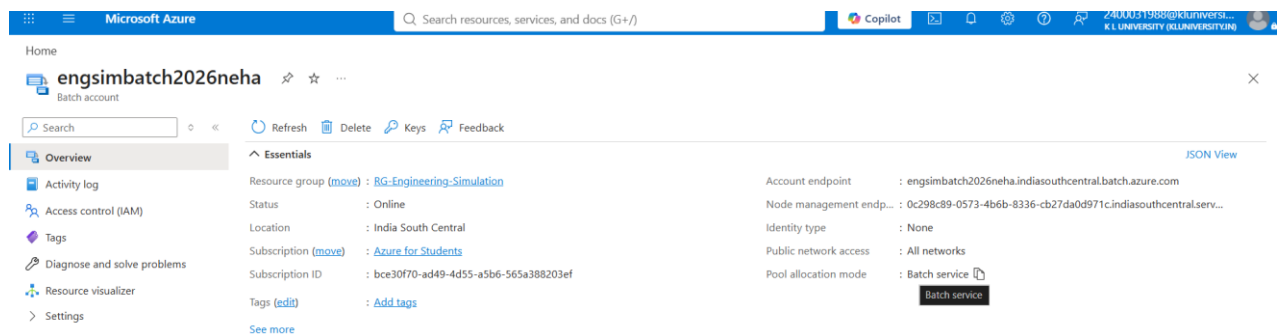
- Replace the demonstration numerical workload with a validated engineering simulation solver.
- Use realistic engineering input files and parameter sets for each scenario.
- Collect task outputs into Azure Storage for centralized result analysis.
- Experiment with larger pools and autoscaling for larger workloads.
- Connect the standalone prototype to the cloud workflow if future project scope requires integration.

19. Appendix: Screenshot Evidence

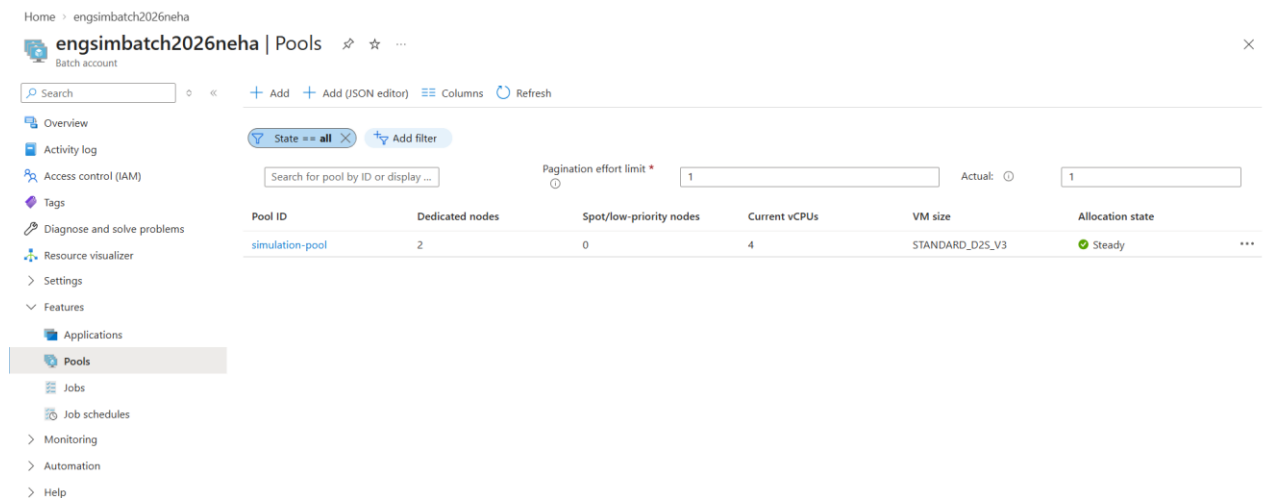
The following evidence is taken from the final presentation and can be used during project review.



Appendix — SimFlow prototype dashboard.



Appendix — Azure Batch account.



Appendix — Azure compute pool.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

2400031988@kkluniversi...
K L UNIVERSITY (KLUNIVERSITY)

Home > engsimbatch2026neha

engsimbatch2026neha | Jobs

Batch account

Search

+ Add + Add (JSON editor) Columns Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Features

Applications

Pools

Jobs

State == all

Add filter

Filter by ID or pool

Pagination effort limit * 1 Actual: 1

ID	State	Pool	Created
simulation-job	Active	simulation-pool	26 Sept 2026, 22:56:55

Appendix — Azure Batch job.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

2400031988@kkluniversi...
K L UNIVERSITY (KLUNIVERSITY)

Home > engsimbatch2026neha | Jobs > simulation-job

simulation-job | Tasks

engsimbatch2026neha

Search

+ Add + Add (JSON editor) Columns Refresh

Overview

General

Properties

Environment settings

Metadata

Tasks

Preparation tasks

Release tasks

Settings

Approximate task count (300 Tasks) 100.00% (300 / 300 Completed)

State == all

Add filter

Filter by task ID (start with)

Pagination effort limit * 1 Actual: 1

Task	State	Created	Exit code
simulation-task-001	Completed	26 Sept 2026, 23:09:02	0
simulation-task-002	Completed	26 Sept 2026, 23:09:02	0
simulation-task-003	Completed	26 Sept 2026, 23:09:02	0
simulation-task-004	Completed	26 Sept 2026, 23:09:02	0
simulation-task-005	Completed	26 Sept 2026, 23:09:02	0
simulation-task-006	Completed	26 Sept 2026, 23:09:02	0
simulation-task-007	Completed	26 Sept 2026, 23:09:02	0
simulation-task-008	Completed	26 Sept 2026, 23:09:02	0
simulation-task-009	Completed	26 Sept 2026, 23:09:02	0
simulation-task-010	Completed	26 Sept 2026, 23:09:02	0
simulation-task-011	Completed	26 Sept 2026, 23:09:02	0

Appendix — Azure task monitoring.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

2400031988@kkluniversi...
K L UNIVERSITY (KLUNIVERSITY)

Home > engsimbatch2026neha | Jobs > simulation-job

simulation-job | Tasks

engsimbatch2026neha

Search

+ Add + Add (JSON editor) Columns Refresh

Overview

General

Properties

Environment settings

Metadata

Tasks

Preparation tasks

Release tasks

Settings

simulation-task-284	Completed	26 Sept 2026, 23:11:50	0
simulation-task-285	Completed	26 Sept 2026, 23:11:50	0
simulation-task-286	Completed	26 Sept 2026, 23:11:50	0
simulation-task-287	Completed	26 Sept 2026, 23:11:50	0
simulation-task-288	Completed	26 Sept 2026, 23:11:50	0
simulation-task-289	Completed	26 Sept 2026, 23:11:50	0
simulation-task-290	Completed	26 Sept 2026, 23:11:50	0
simulation-task-291	Completed	26 Sept 2026, 23:11:50	0
simulation-task-292	Completed	26 Sept 2026, 23:11:50	0
simulation-task-293	Completed	26 Sept 2026, 23:11:50	0
simulation-task-294	Completed	26 Sept 2026, 23:11:50	0
simulation-task-295	Completed	26 Sept 2026, 23:11:50	0
simulation-task-296	Completed	26 Sept 2026, 23:11:50	0
simulation-task-297	Completed	26 Sept 2026, 23:11:50	0
simulation-task-298	Completed	26 Sept 2026, 23:11:50	0
simulation-task-299	Completed	26 Sept 2026, 23:11:50	0
simulation-task-300	Completed	26 Sept 2026, 23:11:50	0

Appendix — Final task results.

References

- Final project presentation: Azure Batch for an Engineering Simulation Workload, Team T177, KLEF, Department of Computer Science and Engineering.
- Azure Portal screenshots and resource configuration evidence included in the final project presentation.